

# **Ch 11: Areas Related to Circles**

**Question Bank | 2M | 3M | 5M**

## **SECTION A - 2 MARKS**

**Q1. Find the area and circumference of a circle of radius 14 cm.**

**Q2. Find the radius of a circle whose circumference is 44 cm.**

**Q3. Find the diameter of a circle whose area is  $154 \text{ cm}^2$ .**

**Q4. If the perimeter and area of a circle are numerically equal, find the radius of the circle. (PYQ)**

**Q5. The radii of two circles are 19 cm and 9 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. (PYQ NCERT)**

**Q6. The radii of two circles are 8 cm and 6 cm. Find the radius of the circle whose area equals the sum of their areas. (PYQ NCERT)**

**Q7. Find the diameter of a circle whose area equals the sum of areas of circles with radii 24 cm and 7 cm.**

**Q8. Find the circumference of a circle whose area is  $6.16 \text{ cm}^2$ .**

**Q9. Find the area of a ring whose outer radius is 20 cm and inner radius is 15 cm.**

**Q10. A square is inscribed in a circle of radius  $r$ . Find the area of the square.**

**Q11. Find the area of a sector of a circle with radius 6 cm and angle  $60^\circ$ . (PYQ NCERT)**

**Q12. Find the area of a sector of a circle with radius 21 cm and angle  $120^\circ$ .**

**Q13. Find the area of a quadrant of a circle whose circumference is 22 cm. (PYQ NCERT)**

**Q14. Find the length of the arc of a circle of radius 12 cm subtending an angle of  $90^\circ$  at the centre.**

**Q15. Find the length of the arc of a circle of radius 21 cm subtending an angle of  $60^\circ$  at the centre.**

**Q16. Find the perimeter of a sector of a circle with radius 21 cm and angle  $120^\circ$ . (PYQ 2019)**

**Q17. Find the perimeter of a sector of radius 10.5 cm and angle  $90^\circ$ .**

**Q18. A sector of a circle of radius 8 cm contains an angle of  $135^\circ$ . Find the area of the sector.**

**Q19. The length of an arc of a circle with radius 12 cm is  $10\pi$  cm. Find the central angle.**

**Q20. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the minor segment. (Use  $\pi = 3.14$ ) (PYQ NCERT)**

**Q21. Find the area of the minor segment of a circle of radius 14 cm when the corresponding sector angle is  $60^\circ$ .**

**Q22. A chord of a circle of radius 14 cm makes a right angle at the centre. Find the areas of both minor and major segments.**

**Q23. The length of the minute hand of a clock is 14 cm. Find the area swept by it in 5 minutes. (PYQ NCERT)**

**Q24. The minute hand of a clock is 12 cm long. Find the area swept by it in 35 minutes. (PYQ 2019)**

**Q25. The hour hand of a clock is 6 cm long. Find the area swept by it from 12:00 noon to 3:00 PM.**

**Q26. Find the number of revolutions made by a circular wheel of area  $1.54 \text{ m}^2$  in rolling a distance of 176 m.**

**Q27. The wheels of a car are of diameter 80 cm. How many complete revolutions does each wheel make in 10 minutes when the car travels at 66 km/hr? (PYQ NCERT)**

**Q28. A bicycle wheel makes 5000 revolutions in travelling 11 km. Find the diameter of the wheel.**

**Q29. A brooch is made with silver wire forming a circle of diameter 35 mm with 5 diameters also made of wire. Find the total length of silver wire required. (PYQ NCERT)**

**Q30. Two circular pieces of equal radii and maximum area, touching each other, are cut from a rectangular card of dimensions  $14\text{ cm} \times 7\text{ cm}$ . Find the area of the remaining piece. (PYQ 2020)**

**Q31. Find the area of the largest circle that can be drawn inside a square of side  $14\text{ cm}$ .**

**Q32. A horse is tied to a peg at a corner of a square field of side  $15\text{ m}$  by a rope of  $5\text{ m}$ . Find the area it can graze. (PYQ NCERT)**

**Q33. Find the perimeter of a semicircle of radius  $7\text{ cm}$ .**

**Q34. Find the area of a sector of radius  $28\text{ cm}$  and central angle  $45^\circ$ .**

**Q35. A square of diagonal  $8\text{ cm}$  is inscribed in a circle. Find the area of the shaded region between the square and the circle.**

**Q36. The wheel of a motorcycle is of radius  $35\text{ cm}$ . How many revolutions per minute must the wheel make to keep a speed of  $66\text{ km/hr}$ ?**

**Q37. Find the area and perimeter of a sector of radius  $14\text{ cm}$  and angle  $45^\circ$ .**

**Q38. Find the difference between the circumferences of two circles whose radii are in ratio  $5:3$  and whose sum of radii is  $32\text{ cm}$ .**

## **SECTION B - 3 MARKS**

**Q1. A chord of a circle of radius 12 cm subtends an angle of  $120^\circ$  at the centre. Find the area of the corresponding segment. (Use  $\pi = 3.14$ ,  $\sqrt{3} = 1.73$ ) (PYQ 2019)**

**Q2. A chord of a circle of radius 15 cm subtends an angle of  $60^\circ$  at the centre. Find the areas of the corresponding minor and major segments. (PYQ 2018)**

**Q3. A chord PQ of length 12 cm subtends an angle of  $120^\circ$  at the centre of a circle. Find the area of the minor segment.**

**Q4. Find the area of the minor segment of a circle of radius 14 cm when the angle of the corresponding sector is  $60^\circ$ . Also find the area of the major segment. (PYQ NCERT)**

**Q5. A chord AB of a circle of radius 10 cm makes a right angle at the centre. Find the area of the major segment. (Use  $\pi = 3.14$ )**

**Q6. In a circle of radius 21 cm, an arc subtends an angle of  $60^\circ$  at the centre. Find the length of the arc, the area of the sector, and the area of the segment. (PYQ NCERT)**

**Q7. The perimeter of a sector of a circle with central angle  $90^\circ$  is 25 cm. Find the area of the minor segment of the circle.**

**Q8. In the figure, ABCD is a square of side 14 cm. With each vertex as centre and radius 7 cm, sectors are drawn inside the square. Find the area of the shaded region. (PYQ 2018 — most repeated)**

**Q9. A square field has side 21 cm. Four cows are tied at the four corners with ropes of 10.5 cm each. Find the area of the field that cannot be grazed. (PYQ 2019)**

**Q10. Four equal circles are described about the four corners of a square so that each circle touches two of the others. If the side of the square is 14 cm, find the area enclosed between the circles.**

**Q11. In the figure, ABCD is a trapezium with  $AB \parallel DC$ ,  $AB = 18$  cm,  $DC = 32$  cm, distance between them = 14 cm. Arcs of equal radii 7 cm with centres A, B, C, D are drawn. Find the area of the shaded region. (PYQ — most repeated)**

**Q12. In the figure, ABCD is a rectangle with  $AB = 14$  cm and  $BC = 7$  cm. With BC as diameter, a semicircle is drawn outside the rectangle. Find the area of the total figure.**

**Q13. Find the area of the shaded region in the figure where two concentric circles of radii 7 cm and 14 cm are drawn and  $\angle AOC = 40^\circ$ . (PYQ 2020)**

**Q14. AB and CD are respectively arcs of two concentric circles of radii 21 cm and 7 cm with centre O. If  $\angle AOB = 30^\circ$ , find the area of the shaded region. (PYQ NCERT)**

**Q15. Two concentric circles of radii 5 cm and 3 cm are drawn. A chord AB of the outer circle touches the inner circle. Find the area of the region between the two circles.**

**Q16. ABC is a quadrant of a circle of radius 14 cm and a semicircle is drawn with BC as diameter. Find the area of the shaded region. (PYQ NCERT)**

**Q17. Find the area of the shaded region where a circular arc of radius 6 cm has been drawn with vertex O of an equilateral triangle OAB of side 12 cm as centre. (PYQ NCERT)**

**Q18. A square OABC is inscribed in a quadrant OPBQ of a circle. If  $OA = 20$  cm, find the area of the shaded region. (Use  $\pi = 3.14$ ) (PYQ NCERT)**

**Q19. OACB is a quadrant of a circle with centre O and radius 7 cm. If  $OD = 4$  cm, find the area of the shaded region. (PYQ NCERT)**

**Q20. Three circles are drawn with vertices of an equilateral triangle of side 4 cm as centres such that each circle touches the other two. Find the area inside the triangle but outside the three circles. (PYQ 2019)**

**Q21. Three semicircles are drawn — one on  $AB = 36$  cm as diameter and one each on halves of AB. Find the area of the shaded region.**

**Q22. A round table cover has six equal designs. If the radius of the cover is 28 cm, find the cost of making the designs at Rs. 0.35 per  $\text{cm}^2$ . (Use  $\sqrt{3} = 1.7$ ) (PYQ NCERT)**

**Q23. An umbrella has 8 equally spaced ribs. If the radius is 45 cm, find the area between two consecutive ribs. (PYQ NCERT)**

**Q24. A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of  $115^\circ$ . Find the total area cleaned at each sweep. (PYQ NCERT)**

**Q25. A lighthouse spreads a red light over a sector of angle  $80^\circ$  to a distance of 16.5 km. Find the area of the sea over which ships are warned. (Use  $\pi = 3.14$ ) (PYQ NCERT)**

**Q26. Find the area of the shaded design in the figure where ABCD is a square of side 10 cm and semicircles are drawn with each side as diameter inside the square. (PYQ 2019)**

**Q27. Calculate the area of the designed region common between two quadrants of circles of radius 8 cm each, in a square ABCD of side 8 cm. (PYQ NCERT)**

**Q28. Four cows are tied with a rope of length 7 m each at the four corners of a square field of side 21 m. Find the area of the field that cannot be grazed by the cows. (PYQ 2019)**

**Q29. A park is in the form of a rectangle  $120\text{ m} \times 100\text{ m}$ . At the centre of the park there is a circular lawn. The area of the park excluding the circular lawn is  $8700\text{ m}^2$ . Find the radius of the circular lawn.**

**Q30. In the figure, PSR, RTQ, and PAQ are three semicircles of diameters 10 cm, 3 cm, and 7 cm respectively. Find the perimeter of the shaded region. (PYQ NCERT)**

**Q31. A path of 4 m width runs around a semicircular grassy plot whose boundary diameter is 16 m. Find the cost of gravelling the path at Rs. 1.50 per square metre.**



**Q32.** Find the area of the shaded region in the figure where arcs of radius 3 cm each are drawn with vertices A, B, C, D of a square ABCD of side 6 cm as centres intersecting each other at mid-points of the sides.

**Q33.** Two flower beds are shown on two sides of a square lawn ABCD of side 56 m. If the centre of each flower bed is the point of intersection of diagonals, find the sum of areas of the lawn and the flower beds. (PYQ NCERT)

**Q34.** A sector of a circle of radius 12 cm has angle  $120^\circ$ . Find the area of the corresponding segment and major segment. (Use  $\pi = 3.14$ ,  $\sqrt{3} = 1.73$ ) (PYQ 2020)

**Q35.** The area of an equilateral triangle ABC is  $49\sqrt{3}$  cm<sup>2</sup>. Taking each vertex as centre, circles are drawn with radius equal to half the length of each side. Find the area of the triangle not covered by the circles.

**Q36.** Find the area of the shaded region where ABCD is a square of side 28 cm and arcs are drawn with each vertex as centre and radius equal to half the side of the square. (PYQ 2020)

**Q37.** In the figure, a circle of radius 7.5 cm is inscribed in a square. Find the area of the shaded region formed between the square and the circle.

**Q38.** A chord AB of a circle of radius 10 cm subtends an angle of  $60^\circ$  at the centre. Find the area of the minor segment. Also find the area of the major segment.

**Q39.** In the figure, ABCD is a trapezium with  $AB \parallel CD$ .  $AB = 18$  cm,  $DC = 32$  cm. A circle is inscribed in the trapezium touching all four sides. If radius is 7 cm, find the area of the trapezium not covered by the circle.

**Q40.** Find the area of the region enclosed between two concentric circles of radii 7 cm and 3.5 cm. Also find what fraction of the outer circle's area this region represents.

**Q41.** An arc of a circle of radius 21 cm subtends an angle of  $60^\circ$  at the centre. Find the area of the corresponding sector and verify by computing the arc length and using the formula  $\text{Area} = \frac{1}{2} \times l \times r$ .

**Q42.** Find the difference of areas of two segments of a circle formed by a chord of length 5 cm subtending an angle of  $90^\circ$  at the centre. (PYQ NCERT Exemplar)

**Q43.** In the figure, arcs are drawn with centres A, B, C, D of a square ABCD of side 12 cm intersecting at midpoints P, Q, R, S of the sides. Find the area of the shaded region. (Use  $\pi = 3.14$ ) (PYQ NCERT Exemplar)

### **SECTION C - 5 MARKS**

**Q1.** In the given figure, ABCD is a square of side 14 cm. With each vertex as centre, a circle is drawn such that each circle

touches two of the other circles externally. Find the area of the shaded region. *(PYQ 2018 — most repeated)*

**Q2.** In the figure, four squares are drawn with vertices of a square of side 10 cm. Find the area of the shaded region. *(PYQ 2019)*

**Q3.** Find the area of the shaded region in the figure where ABCD is a square of side 14 cm and four semicircles are drawn with AB, BC, CD, and DA as diameters inside the square. *(PYQ 2020)*

**Q4.** In the figure, ABC is a right triangle with  $\angle ABC = 90^\circ$ , BC = 6 cm and AB = 8 cm. A circle with centre O and radius r is inscribed in the triangle. Find the radius r and the area of the shaded region. *(PYQ 2019)*

**Q5.** In the figure, O is the centre of a circle such that diameter AB = 13 cm and AC = 12 cm. BC is joined. Find the area of the shaded region. (Use  $\pi = 3.14$ ) *(PYQ 2018)*

**Q6.** In the figure, ABCD is a trapezium with  $AB \parallel CD$ , AB = 18 cm, DC = 32 cm and distance = 14 cm. Arcs of radius 7 cm are drawn with centres A, B, C, D. Find the area of the shaded region. *(PYQ — frequently asked)*

**Q7.** In the figure, a square OABC is inscribed in a quadrant OPBQ. If OA = 20 cm, find the area of the shaded region. (Use  $\pi = 3.14$ ) *(PYQ NCERT)*

**Q8. A sector of a circle of radius 12 cm has an angle of  $120^\circ$ . Find the area of the corresponding segment. Also find the area of the major segment. (Use  $\pi = 3.14$ ,  $\sqrt{3} = 1.73$ ) (PYQ 2020)**

**Q9. Find the area of the shaded region in the figure enclosed between two concentric circles of radii 7 cm and 14 cm where  $\angle AOC = 40^\circ$ . Also find the perimeter of the shaded region. (PYQ 2020)**

**Q10. ABC is a quadrant of a circle of radius 14 cm and a semicircle is drawn with BC as diameter. Find the area of the shaded region. Verify your answer by an alternative method. (PYQ NCERT)**

**Q11. In the figure, two concentric circles of radii  $a$  and  $b$  ( $a > b$ ) are given. A chord AB of the larger circle touches the smaller circle at C. Find the area of the shaded region between the two circles in terms of AB. (PYQ)**

**Q12. Case Study: A pookalam is a flower arrangement made in circular rings of alternate colours. Rings of flowers are placed at radii 5 cm, 10 cm, 15 cm, and 20 cm from the centre. Find: (i) area of innermost circle (ii) area of each ring (iii) total area of the pookalam (iv) area of the outermost ring as a fraction of total area. (PYQ 2023 case-based)**

**Q13. The area of an equilateral triangle ABC is  $17320.5 \text{ cm}^2$ . With each vertex as centre and radius equal to half the length of each side, circles are drawn. Find the area of the shaded region. (Use  $\pi = 3.14$ ,  $\sqrt{3} = 1.73205$ ) (PYQ NCERT)**

**Q14.** On a square handkerchief, nine circular designs each of radius 7 cm are made. Find the area of the remaining portion of the handkerchief. *(PYQ NCERT)*

**Q15.** In a circular table cover of radius 32 cm, a design is formed leaving an equilateral triangle ABC in the middle. Find the area of the design. *(PYQ NCERT)*

**Q16.** Four equal circles, each of radius 5 cm, touch each other as shown in the figure. Find the area of the region between them. *(PYQ)*

**Q17.** A chord PQ of length 12 cm subtends an angle of  $120^\circ$  at the centre of a circle. Find the area of the minor segment cut off by the chord PQ. Also find the area of the major segment and verify that their sum equals the area of the complete circle.

**Q18.** Case Study: A garden is designed in the shape of a rectangle PQRS with  $PQ = 30$  m and  $QR = 20$  m. Semicircles are drawn on each shorter side. A horse is tied at the midpoint M of PQ with a rope of 14 m. Find: (i) area grazeable inside the garden (ii) area grazeable outside (iii) total area grazeable (iv) area of the garden not grazed. *(PYQ 2024 case-based)*

**Q19.** A footpath of uniform width runs all around the inside of a rectangular field 54 m long and 35 m wide. If the area of the path is  $420 \text{ m}^2$ , find the width of the path. Also find the area of the inner rectangle.

**Q20.** The area of an equilateral triangle is  $49\sqrt{3} \text{ cm}^2$ . Taking each angular point as centre, circles are drawn with radius

equal to half the length of the side of the triangle. Find the area of the triangle not covered by the circles. *(PYQ 2019)*

**Q21.** In the figure, a decorative block is made of two solids. The block has a circular base of radius 3.5 cm and total height 6.5 cm. A cone of height 2.5 cm is surmounted on a cylinder of height 4 cm. Find: (i) the area of the circular base (ii) the curved surface area of the cylindrical part (iii) the curved surface area of the conical part (iv) total surface area of the block.

*(Cross-chapter HOTS — surface area meets circles)*

**Q22.** Find the area of the shaded region in the figure, if ABCD is a square of side 28 cm, and arcs are drawn with each vertex as centre and radius equal to half the side of the square. Also find the perimeter of the shaded region. *(PYQ 2020)*

**Q23.** Case Study: An architect designs a circular park of radius 21 m. Inside the park, an equilateral triangular path is made. Flower beds in the shape of segments are made between the triangle and the circle. Find: (i) the side of the equilateral triangle (ii) the area of the triangle (iii) the area of each flower bed segment (iv) the total area of all three flower beds (v) the cost of planting flowers at Rs. 50 per  $\text{m}^2$ . *(PYQ 2024 case-based)*

**Q24.** Two circles touch externally. The sum of their areas is  $130\pi \text{ cm}^2$  and the distance between their centres is 14 cm. Find the radii of the two circles. Also find the perimeter of the figure formed by the two circles.

**Q25.** In the figure, PSR, RTQ, and PAQ are three semicircles of diameters 10 cm, 3 cm, and 7 cm respectively. Find the area of

the shaded region. Also find the perimeter of the shaded region.  
*(PYQ NCERT)*

**Q26.** A circular disc of radius 6 cm is divided into three sectors with central angles  $90^\circ$ ,  $120^\circ$ , and  $150^\circ$ . Find the ratio of the areas of three sectors. Also find the ratio of arc lengths of the three sectors. *(PYQ NCERT Exemplar)*

**Q27.** The minute hand of a clock is 12 cm long. Find the area swept by the minute hand between 9:00 AM and 9:35 AM. Also find the distance travelled by the tip of the minute hand in this duration.